

NYC Airbnb Market Segmentation Analysis

Executive Summary

This report presents a K-Means clustering analysis of 20,590 NYC Airbnb listings. By evaluating 10 numeric features through a KNIME workflow, we identified three distinct market segments: Long-stay Commercial, Premium/Luxury, and Popular Budget. The optimal cluster count ($k=3$) was determined using the Elbow Method and Silhouette Score (0.2975).

Methodology & Workflow

The analysis was conducted using a 8-node KNIME workflow. The process included data ingestion, feature selection, and normalization to ensure balanced clustering weights.

- File Reader: Data ingestion.
- Column Filter: Selection of 10 numeric features.
- Missing Value: Data cleaning and imputation.
- Normalizer (Z-score): Scaling features to mean=0 and std_dev=1.
- k-Means ($k=3$): Core clustering algorithm execution.
- Color Manager: Visual distinction for clusters.
- Scatter/Parallel Coordinates Plot: Visual validation.
- GroupBy statistics: Aggregate summary for profiling.

Market Segment Profiles

Cluster 0: Long-stay Commercial

Representing 16.8% of listings ($n=3,454$), this segment is dominated by commercial operators. Key characteristics include:

- Average Price: \$145
- Minimum Nights: 35.1 nights (high-duration rentals)
- Host Profile: Professional operators with an average of 68.5 listings each.
- Geography: Primarily concentrated in Manhattan.
- Note: This cluster shows extremely low ratings (0.11), suggesting dormant or purely corporate-managed listings.

Cluster 1: Premium / Luxury

Representing 13.6% of listings ($n=2,808$), this is the high-end tier of the market.

- Average Price: \$331 (highest across all segments)
- Property Size: Larger homes averaging 2.79 bedrooms and 3.67 beds.
- Host Profile: Small, independent hosts (avg. 7.2 listings).
- Customer Satisfaction: Solid rating of 4.15/5.0.
- Geography: Spread evenly across Brooklyn and Manhattan.

Cluster 2: Popular Budget

The dominant cluster, comprising 69.6% of the market (n=14,328). These are the 'bread-and-butter' listings.

- Average Price: \$132 (most affordable).
- Property Size: Small (1 bedroom units).
- Engagement: Highly reviewed (53.5 reviews avg) with an outstanding 4.74 rating.
- Host Profile: Mostly independent hosts (avg. 9.3 listings).
- Geography: Present across all NYC boroughs.

Borough Distribution Data

Borough	C0 Long-stay	C1 Premium	C2 Budget	Total
Bronx	129	97	722	948
Brooklyn	1115	1171	5411	7697
Manhattan	1643	964	5293	7900
Queens	538	529	2687	3754
Staten Island	29	47	215	291

Conclusion

The analysis reveals a highly fragmented market. While the Budget segment (C2) drives the majority of volume and guest satisfaction, the Luxury segment (C1) commands the highest margins. The Commercial segment (C0) represents a specific niche of high-minimum-stay properties that operate differently from traditional short-term rentals.